

Quantitative Aptitude: HCF and LCM

Comprehensive Study Notes, 12 Solved Examples & 25 Practice Questions with Answer Key

1. Theory & Fundamental Concepts of HCF and LCM

Highest Common Factor (HCF) and Least Common Multiple (LCM) form an integral part of Quantitative Aptitude. They are heavily applied in number problems, traffic signal synchronizations, circular tracks, and fraction simplifications.

Core Definitions:

- **Factors:** A number that divides another number completely without leaving any remainder is called a factor. (e.g., factors of 12 are 1, 2, 3, 4, 6, 12).
- **Multiples:** A number obtained by multiplying a given number by an integer is a multiple. (e.g., multiples of 4 are 4, 8, 12, 16, \dots).
- **Highest Common Factor (HCF / GCD):** The HCF of two or more numbers is the greatest number that divides each of them exactly. Also known as Greatest Common Divisor (GCD).
- **Least Common Multiple (LCM):** The LCM of two or more numbers is the lowest/smallest number that is divisible by each of them exactly.

Important Formulas & Properties:

- For any two numbers a and b : $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$
- HCF of given numbers is always a factor of their LCM.
- $\text{HCF of Fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$
- $\text{LCM of Fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$
- Co-prime numbers always have an HCF of 1. Their LCM is equal to their product.

2. Solved Examples (12 Problems)

Example 1: Find the HCF of 24, 36, and 60.

Solution: Prime factorizations:

$$24 = 2^3 \times 3, 36 = 2^2 \times 3^2, 60 = 2^2 \times 3 \times 5$$

HCF is the product of smallest powers of common prime factors: $2^2 \times 3 = 4 \times 3 = \mathbf{12}$.

Example 2: Find the LCM of 12, 15, 20, and 27.

Solution: Prime factorizations:

$$12 = 2^2 \times 3, 15 = 3 \times 5, 20 = 2^2 \times 5, 27 = 3^3$$

LCM is the product of highest powers of all prime factors: $2^2 \times 3^3 \times 5 = 4 \times 27 \times 5 = \mathbf{540}$.

Example 3: Find the HCF and LCM of fractions $\frac{2}{3}$, $\frac{8}{9}$, $\frac{16}{81}$, $\frac{10}{27}$.

Solution:

$$\text{HCF} = \frac{\text{HCF}(2, 8, 16, 10)}{\text{LCM}(3, 9, 81, 27)} = \frac{2}{81}$$

$$\text{LCM} = \frac{\text{LCM}(2, 8, 16, 10)}{\text{HCF}(3, 9, 81, 27)} = \frac{80}{3}$$

Example 4: The HCF of two numbers is 11 and their LCM is 770. If one number is 275, find the other number.

Solution: Using formula: $\text{HCF} \times \text{LCM} = \text{First Number} \times \text{Second Number}$
 $11 \times 770 = 275 \times x \implies x = \frac{11 \times 770}{275} = \frac{8470}{275} = \mathbf{308}$.

Example 5: Find the greatest number that will divide 43, 91, and 183 so as to leave the same remainder in each case.

Solution: The required number is the HCF of differences of numbers taken in pairs:

$$\text{Differences: } (91 - 43) = 48, (183 - 91) = 92, (183 - 43) = 140.$$

HCF of 48, 92, and 140 is $\mathbf{4}$.

Example 6: Find the largest number of four digits which is exactly divisible by 12, 15, 18, and 27.

Solution: First, find LCM of 12, 15, 18, and 27 = 540.

The greatest 4-digit number is 9999. Divide 9999 by 540, remainder is 279.

$$\text{Required number} = 9999 - 279 = \mathbf{9720}.$$

Example 7: Find the smallest number which when divided by 20, 25, 35, and 40 leaves remainders 14, 19, 29, and 34 respectively.

Solution: Note that (Divisor - Remainder) is constant in each case: $20 - 14 = 6$, $25 - 19 = 6$, etc.
LCM of 20, 25, 35, 40 = 1400.
Required number = $\text{LCM} - \text{Constant Difference} = 1400 - 6 = \mathbf{1394}$.

Example 8: Three bells toll together at intervals of 9, 12, and 15 minutes respectively. If they toll together now, after how much time will they toll together next?

Solution: The time interval when they toll together again is the LCM of 9, 12, and 15.
LCM = $\mathbf{180}$ minutes (or 3 hours).

Example 9: The ratio of two numbers is 3:4 and their HCF is 4. Find their LCM.

Solution: Let numbers be $3x$ and $4x$. Since HCF is 4, numbers are 12 and 16.
LCM = Product of numbers / HCF = $\frac{12 \times 16}{4} = \mathbf{48}$ (or $3 \times 4 \times 4 = 48$).

Example 10: Find the least multiple of 13 which when divided by 4, 5, 6, 7, and 8 leaves remainder 2 in each case.

Solution: LCM of 4, 5, 6, 7, 8 is 840.
Number is of the form $840k + 2$. For this to be a multiple of 13, test values of k .
For $k = 2$, $840(2) + 2 = 1682$ (not divisible by 13). For $k = 3$, $840(3) + 2 = 2522$ ($2522 \div 13 = 194$).
Thus, $\mathbf{2522}$.

Example 11: Find the greatest number that divides 2011 and 2423 leaving remainders 9 and 5 respectively.

Solution: Required number is the HCF of $(2011 - 9)$ and $(2423 - 5)$.
HCF of 2002 and 2418 = $\mathbf{26}$.

Example 12: Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers.

Solution: Numbers are 15×13 and 11×13 .
Numbers are $\mathbf{195}$ and $\mathbf{143}$.

3. Practice Questions (25 Questions)

Q1. Find the HCF of 108, 288, and 360.

- (A) 12 (B) 24
(C) 36 (D) 54
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Q2. Find the LCM of 72, 108, and 210.

- (A) 3780 (B) 1512
(C) 2520 (D) 5040
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Q3. Find the HCF of fractions $\frac{9}{10}$, $\frac{12}{25}$, $\frac{18}{35}$, $\frac{21}{40}$.

- (A) $\frac{3}{1400}$ (B) $\frac{9}{1400}$
(C) $\frac{3}{2800}$ (D) $\frac{27}{1400}$
-

Q4. The LCM of two numbers is 48. The numbers are in the ratio 2:3. Find the sum of the numbers.

- (A) 28 (B) 32
(C) 40 (D) 60
-

Q5. Product of two co-prime numbers is 117. Their LCM is:

- (A) 9 (B) 13
(C) 39 (D) 117
-

Q6. The HCF of two numbers is 23 and the other two factors of their LCM are 13 and 14. Find the larger number.

- (A) 299 (B) 322
(C) 276 (D) 345
-

Q7. Find the greatest number that will divide 55, 127, and 175, leaving the same remainder in each case.

- (A) 12 (B) 16
(C) 24 (D) 28
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Q8. Find the least number which when divided by 12, 16, 24, and 36 leaves remainders 7, 11, 19, and 31 respectively.

- (A) 139 (B) 149
(C) 143 (D) 155
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Q9. Four bells ring at intervals of 5, 6, 8, and 9 seconds respectively. All rings together at some time. After how many minutes will they ring together next?

- (A) 6 minutes
(C) 18 minutes

- (B) 12 minutes
(D) 24 minutes

Q10. The HCF of two numbers is 12 and their difference is 12. Which of the following can be the numbers?

- (A) 66, 54
(C) 72, 84

- (B) 84, 96
(D) All of these

Q11. Find the greatest 5-digit number which is divisible by 15, 20, 24, and 32.

- (A) 99840
(C) 99900

- (B) 99600
(D) 99720

Q12. Three numbers are in the ratio **1:2:3** and their HCF is 12. The numbers are:

- (A) 12, 24, 36
(C) 5, 10, 15

- (B) 24, 48, 72
(D) 10, 20, 30

Q13. If **$ab = 432$** and **$\text{ext}\{HCF\}(a, b) = 6$** , find **$\text{ext}\{LCM\}(a, b)$** .

- (A) 36
(C) 108

- (B) 72
(D) 144

Q14. Find the least number which when divided by 8, 12, 18, and 24 leaves a remainder 5 in each case.

- (A) 72
(C) 67

- (B) 77
(D) 83

Q15. The sum of two numbers is 216 and their HCF is 27. How many such pairs of numbers can exist?

- (A) 1
(C) 3

- (B) 2
(D) 4

Q16. Find the LCM of **$2^3 \times 3^2 \times 5$** , **$2^2 \times 3^3 \times 7$** , and **$2 \times 3 \times 11$** .

- (A) 27720
(C) 6930

- (B) 13860
(D) 41580

Q17. Find the maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets the same number of pens and pencils.

- (A) 91 (B) 101
(C) 110 (D) 143
-

Q18. What is the least number which when increased by 5 is completely divisible by 24, 32, 36, and 64?

- (A) 571 (B) 576
(C) 581 (D) 283
-

Q19. The ratio of two numbers is **4:5** and their LCM is 140. Find the numbers.

- (A) 24, 30 (B) 28, 35
(C) 32, 40 (D) 36, 45
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Q20.F Find the HCF of **1.2** and **0.36**.

- (A) 1.2 (B) 0.12
(C) 0.036 (D) 0.6
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Q21. LCM of two numbers is 12 times their HCF. The sum of HCF and LCM is 403. If one number is 93, find the other number.

- (A) 124 (B) 134
(C) 148 (D) 156
-

Q22. The LCM of two co-prime numbers is 551. Find their HCF.

- (A) 19 (B) 29
(C) 1
(D) Cannot be determined
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Q23. Find the smallest 4-digit number which is divisible by 18, 24, and 32.

- (A) 1036 (B) 1152
(C) 1008 (D) 1224
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Q24. Three runners running around a circular track of 12 km length complete a round in 3, 4, and 6 hours respectively. When will they meet again at the starting point?

- (A) After 6 hours (B) After 12 hours
(C) After 18 hours (D) After 24 hours
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Q25. Find the greatest number that will divide 400, 445, and 535 leaving remainders 4, 5, and 7 respectively.

(A) 16

(B) 24

(C) 32

(D) 36

4. Answer Key

Correct answers for all 25 practice questions:

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(C) 36	Q8	(C) 143	Q15	(B) 2	Q22	(C) 1
Q2	(A) 3780	Q9	(B) 12 minutes	Q16	(A) 27720	Q23	(C) 1008
Q3	(A) 1400	Q10	(D) All of these	Q17	(A) 91	Q24	(B) After 12 hours
Q4	(B) 32	Q11	(A) 99840	Q18	(A) 571	Q25	(A) 16
Q5	(D) 117	Q12	(A) 12, 24, 36	Q19	(B) 28, 35		
Q6	(B) 322	Q13	(B) 72	Q20	(B) 0.12		
Q7	(D) 28	Q14	(B) 77	Q21	(A) 124		